AI & DATA SCIENCE · LECTURE

Why Normalize?

The theory behind feature scaling in linear and logistic regression

we normalize the data before we train it to have the range as close as possible like 5 to 10 and we standardize to make the data as close to the median

Companion to the hands-on notebook

THE PROBLEM

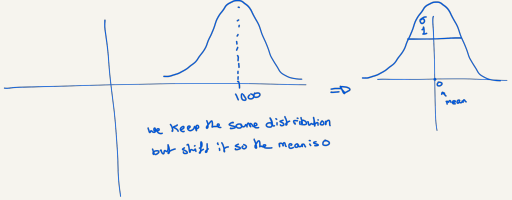
Features rarely share a scale

Feature	Std. deviation
worst area	569.4
mean area	351.9
area error	45.5
smoothness error	0.0030
fractal dimension error	0.0026

Breast cancer dataset, 30 features: the three largest and the two smallest standard deviations.

200,000×

spread between the largest and smallest standard deviation among the 30 features



THE PROBLEM

Two standard ways to rescale

Standardization

new x ↓ mean ↓ standard deviation ↓

$$z = (x - \mu) / \sigma$$

Each feature gets mean 0 and standard deviation 1. The default choice for regression models.

Min-max scaling

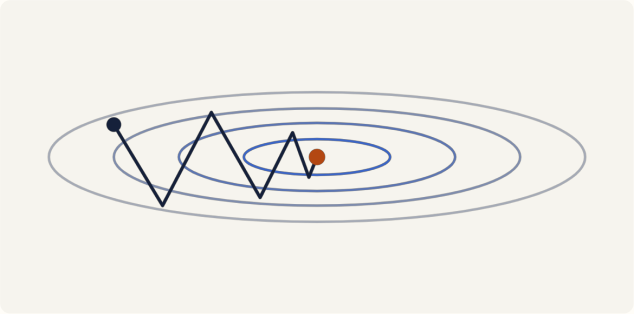
$$x' = (x - \min) / (\max - \min)$$

Each feature is mapped into the range [0, 1]. Sensitive to outliers.

Rule: learn μ, σ, min and max from the training set only, then apply them to every other set.

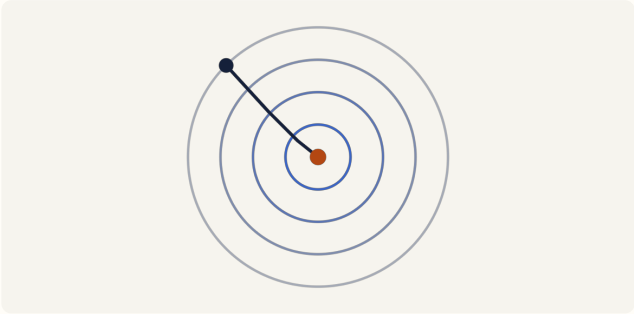
LINEAR REGRESSION

The geometry of the loss surface



Raw features: a long, narrow valley. Gradient descent zigzags and needs a tiny η.

how many steps needed to converge



Scaled features: a round bowl. The gradient points at the minimum.

less steps are needed to convert for the scaled data

LOGISTIC REGRESSION

What happens in practice

2,342 → 19

iterations for the solver to converge (raw → scaled)

less loss

0.951 → 0.986

test accuracy with the default limit of 100 iterations

higher accuracy

linear regression, accuracy isn't affected

logistic regression, accuracy could be affected

2.40 → 0.127

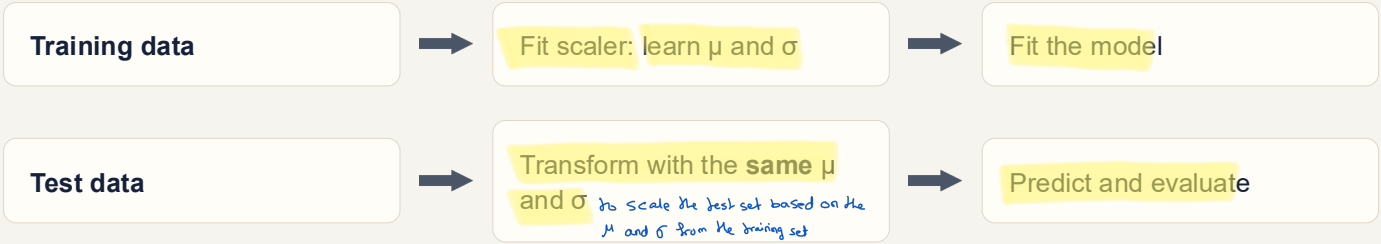
training log-loss after 500 gradient steps at η = 0.01

0.902 → 0.958

test accuracy of the from-scratch gradient descent

IN PRACTICE

Fit the scaler on training data only



Scaling the full dataset first lets test rows influence μ and σ. Information leaks into training and the score is too optimistic. A scikit-learn Pipeline applies this rule automatically in every cross-validation fold.

IN PRACTICE

Choosing a scaler

Scaler	Statistics used	Outlier sensitivity
StandardScaler	mean and standard deviation	Moderate
MinMaxScaler	minimum and maximum	High
RobustScaler	median and IQR	Low

Example with one salary of 1,000,000 among five ordinary salaries: min-max squeezes the five into 0 to 0.03. Robust scaling keeps them spread from −0.88 to 0.64.